

Ratio Correlation Fixes for the Fiscal Health Index

2026-09-16 · @Someone

Executive summary

The problem. The fiscal health index factors a correlation matrix of 14 financial ratios built from about 20 Form 990 lines. Ratios that share a line correlate even when nothing links the organizations, and shares of one total correlate negatively by definition. So the "dimensions" the index discovers may be arithmetic rather than evidence — and nothing in standard model output reveals which.

The solution. Judge every correlation against what the arithmetic alone would produce. Shuffle the 990 lines across organizations of similar size, rebuild the ratios through the identical pipeline, and read the result as the baseline. Whatever exceeds that baseline is a finding; whatever matches it is construction. Relationships that survived were then written as pre-specified hypotheses on dollar lines and tested on a six-year panel, and the index itself was validated against organizations that later stopped filing.

What was built. A 2019–2024 panel of 356,288 organizations filing every year, a companion cohort of 280,158 that preserves exits, and nine analysis documents in `dev/health-index/`.

Key findings

- **Most of the correlation structure is construction.** 63 of 91 ratio pairs share a 990 line. Shuffled data reproduce the four dimensions' loading patterns at congruence 0.97–0.99, and parallel analysis against that baseline keeps 1 component rather than 5.
- **No transformation fixes it.** Pairwise correction breaks the matrix, size adjustment erases `assets_rev`, log-ratios are undefined for 42–69% of filers, and redesigning the variable set only moves the artifact into composition links. The baseline is the fix, not a repair.
- **The financing relationships are real but small.** Long-term debt tracks fixed assets and short-term liabilities track current assets, in every year and inside organizations. But only 7–11 cents of each new dollar of property arrives as debt, against a cross-sectional elasticity near 0.48: the cross-section is mostly sorting organizations into types.

- **A surplus dollar becomes 37 cents of cash, 32 of investments, 14 of property, and 12 of debt repayment.** Deficits are absorbed by cash (58 cents), not borrowing. The accounting identity checks these estimates exactly.
- **Earned revenue and fundraising trade off across organizations, not within them.** Organizations that lost earned revenue in the pandemic did not raise fundraising afterwards. These ratios describe a business model, not a management lever.
- **Equal weighting hurts.** The equal-weighted composite predicts operating distress worse (AUC 0.565) than the performance dimension alone (0.621).

Evidence the index works

- **It predicts organizational failure.** Against a confirmed two-year filing gap, the three dimensions plus size reach **AUC 0.715**. Among organizations above \$1m in expenses, where a gap is unlikely to mean a drop to the 990-N postcard, the reweighted index reaches **0.642 against 0.554 for size alone** — the clearest evidence in this work that the measurement model carries real information.
- **Triage lift is roughly threefold.** The bottom decile of the reweighted index contains organizations with a 15.7–16.5% filing-gap rate, against 5.4% overall.
- **The dimensions divide the work.** Performance predicts deficits (AUC 0.62) and net-asset declines (0.58); reserves predicts failure and cash outcomes. Each adds information the other lacks, so both belong in the index.
- **The structure is stable.** Cross-sectional coefficients hold across 2019–2023 and across random halves, size groups, transformations, and an independent year, with only a visible dip in 2020 traceable to pandemic relief loans.

Recommendations

- Add a construction-baseline step before factoring: audit which pairs share lines, and use the shuffled baseline for retention decisions and for reading loadings.
- Replace equal weights with roughly **55–60% performance and 40–45% reserves**; report leverage and asset structure as context, since neither earns predictive weight.
- Present results as a short dashboard with size alongside the rank, not as a single number.
- Do not add the matching-gap indicator: it is a persistent financing posture with almost no predictive value.

Limits

Filing gaps are not the same as dissolution; small organizations can drop to the 990-N postcard, and the 2024 file is still incomplete. All estimates are associations. The next steps are separating true exits from postcard downshifts using IRS status data, and re-fitting the weights against the failure outcome rather than the deficit proxies.

Coupling shifts the baseline; closure removes information

Coupling and closure both manufacture correlation, but only closure limits what the correlation can be. That difference decides which fixes can work.

Coupling. In logs the ratio algebra is exact. With $a = \log A$, $b = \log B$, $c = \log C$:

$$\text{cov}(a - c, b - c) = \sigma_{ab} - \sigma_{ac} - \sigma_{bc} + \sigma_c^2$$

If A and B are unrelated to C and all three have equal variance, the ratio correlation is $(\rho_{ab} + 1) / 2$. Its attainable range is $[0, 1]$, so a ratio correlation of 0.5 means uncorrelated components. Every term is observable on the 990, so undoing coupling amounts to returning to the components; no pair-by-pair rescaling is needed.

Without logs, the law of total covariance puts all of the coupling in one term:

$$\text{cov}(X, Y) = E[\text{cov}(X, Y \mid C)] + \text{cov}(E[X \mid C], E[Y \mid C])$$

Residualizing each ratio on a flexible function of its denominator deletes the second term. If the numerators are independent given C, the remaining correlation is exactly zero. This works with signed numerators, zeros, and asinh or rank transforms.

Closure. Shares of one whole have covariance rows that sum to zero. For three shares, $\text{cov}(s_1, s_2) = (v_3 - v_1 - v_2) / 2$, so the correlations are fully determined by the variances. No rescaling recovers information that is not there. Log-ratio coordinates (CLR, ILR) are the established repair, but they need strictly positive parts.

Three limits apply to every fix.

- **Artifact or construct depends on the pair, not the arithmetic.** Total expenses in the denominator of `months_cash` and `liquid_months` is part of what both measure: resources per month of spending. The same shared term linking `months_cash` and `admin_exp` is pure artifact.

- **The null depends on the size reference.** A shared denominator creates artifact only to the extent it varies independently of organization size. Take expenses as the size reference and expense coupling vanishes by definition; take assets and it returns.
- **Pairwise corrections break the matrix.** Adjusting 91 correlations one at a time generally yields negative eigenvalues. A usable fix transforms variables, not correlations, so the matrix stays valid for PCA and FA.

The 14 ratios as raw 990 lines

The 14 ratios are built from about 20 lines. Three accounting identities make them share far more than their surface formulas suggest. Definitions follow `dev/health-index/build-ratio-frame.R`. Eleven ratios enter the four-dimension model. `self_suff`, `admin_exp`, and `fundr_exp` are the standalone measures forced into the five-component table.

Ratio	Formula	Numerator lines	Denominator lines	Dimension (5-PC table)
<code>current</code>	ca / cl	cash, recv, othca	ap, grant payable	Reserves; Leverage (-)
<code>months_cash</code>	$12 (\text{cash} + \text{recv}) / (E - \text{dep})$	cash, recv	E, dep	Reserves
<code>liquid_months</code>	$12 (\text{una} - \text{land} + \text{mtg}) / E$	UNA, land (-), mtg	E	Reserves
<code>cash_assets</code>	cash / A	cash	A	Asset structure
<code>debt_assets</code>	L / A	L (incl. ap, mtg)	A	Leverage
<code>payables_una</code>	$\text{ap} / (\text{ap} + \text{una}^*)$	ap	ap, UNA	Leverage
<code>surplus_margin</code>	$(R - E) / \max(R, E)$	R, E (-)	R or E	Performance
<code>return_assets</code>	$(R - E) / A$	R, E (-)	A	Performance

Ratio	Formula	Numerator lines	Denominator lines	Dimension (5-PC table)
na_growth	$\Delta NA / \text{mean } NA $	NA_eoy, NA_boj (-)	NA_eoy, NA_boj	Performance
self_suff	progrev / E	progrev (part of R)	E	Standalone
admin_exp	$M / (P + M + Fu)$	M	P, M, Fu (= E)	Standalone
fundr_exp	$Fu / (P + M + Fu)$	Fu	P, M, Fu (= E)	Standalone
land_assets	land / A	land	A	Asset structure (-)
assets_rev	A / R	A	R (E if $R \leq 0$)	Reserves; Asset structure (-)

The identities that link them.

- **Assets are a sum.** $A = \text{cash} + \text{recv} + \text{othca} + \text{land} + \text{other}$, so four ratios place a part of A in a numerator while four others divide by A.
- **Net assets are a residual.** $NA = A - L$ and $\text{una} \subset NA$, so debt_assets, payables_una, liquid_months, and na_growth all move with the same balance sheet gap.
- **Flows nearly coincide.** Functional expenses sum to E, revenue tracks expenses closely, and $\Delta NA \approx R - E$. One flow scale sits behind seven denominators and three numerators.

assets_rev is the ratio of the two scale variables themselves. It is the one variable that any size-based fix will erase.

Pairwise audit: 63 of 91 pairs share a line

63 of the 91 pairs are linked by construction. In 48 the shared line is outside what either ratio measures, in 8 it is the construct itself, and 7 are mixed. The formulas predicted the direction of the simulated null for 28 of 32 pairs where it reached $|r| \geq 0.05$. The null also exposed 4 links the formulas missed, all running through unrestricted net assets or total assets; they are reclassified below. The full table is in the Pairwise audit tab.

Link type	Pairs	Forced sign	Example	Artifact / construct / mixed	Mean r observed	Mean r null (flow / stock)	Fix that fits the pair alone
Shared numerator	10	+	current × cash_assets (cash)	4 / 4 / 2	0.34	0.37 / 0.31	Keep, combine, or benchmark
Closure	2	–	cash_assets × land_assets	2 / 0 / 0	0.29	0.27 / 0.28	Log-ratio contrast
Identity	16	+ or –	debt_assets × na_growth (NA = A – L)	10 / 3 / 3	0.25	0.27 / 0.23	Benchmark only
Reciprocal	11	–	land_assets × assets_rev (A)	10 / 0 / 1	0.24	0.25 / 0.19	Benchmark only
Shared denominator	24	+	months_cash × admin_exp (E)	22 / 1 / 1	0.13	0.04 / 0.04	Residualize on E (18) or A (5)
None	28	0	self_suff × debt_assets	—	0.08	0.02 / 0.01	None needed

Classic shared-denominator coupling is the weakest artifact in this set. Coupling needs a denominator that varies independently of the numerator. Most shared-E pairs put a part of E, or a flow that scales with E, on top (functional expenses, program revenue, the surplus), and their construction correlation is zero. Shared-E coupling survives only where a stock sits over a flow: `months_cash`, `liquid_months`, `assets_rev`. The strong artifacts are shared numerators, closure, identities, and reciprocal links.

What the audit already says about the model.

- **Every dimension is anchored by a construction link.** Performance ratios share R – E; leverage ratios share payables and the net-asset residual; asset structure is a closure pair; reserves share E.
- **The cross-loadings in the five-component table are the artifact pairs.** `current` on leverage (–0.41) is the reciprocal payables link. `admin_exp` on reserves (0.43) is the shared E denominator. `assets_rev` on asset structure (–0.49) is the reciprocal A link.
- **Only 5 of 63 linked pairs call for no fix**, and 28 have no transformation that removes the link without changing the ratio's meaning. Those need a benchmark, not a repair.

No single fix is internally consistent

No transformation removes construction correlation across all 91 pairs while keeping the matrix valid and all 14 ratios intact. Each candidate repairs one link type, leaves the others alone, and often breaks something the others need. Figures below are from the 2023 simulation.

Fix	Link types it repairs	Mean null r after (flow anchor; 0.139 before)	Valid matrix (min eigenvalue)	Cost
F1 Subtract null correlation pair by pair	All, in principle	~0 by construction	No (-0.23)	Nearest valid matrix scrambles the structure; congruence with as-built 0.36–0.74 on four of five components
F2 Residualize every ratio on log E and log A splines	Shared denominator (0.042 → 0.012), partly reciprocal and identity	0.103	Yes (0.20)	Erases 73% of <code>assets_rev</code> ; reserves and asset structure rebuilt (congruence 0.38, 0.47); closure and shared numerators untouched
F3 Residualize each ratio on its own denominator's scale	Shared denominator, weakly	0.129	Yes (0.17)	Structure intact (congruence ≥ 0.95), but removes little

Fix	Link types it repairs	Mean null r after (flow anchor; 0.139 before)	Valid matrix (min eigenvalue)	Cost
Log-ratio contrasts for closed blocks (not simulated)	Closure only (2 pairs)	—	Yes	Undefined for 41.5% of filers (cash or land ≤ 0) and 68.6% (admin or fundraising ≤ 0)
Combine shared-numerator ratios	Shared numerator (3 performance pairs)	—	Yes	Removes the performance dimension as a separate test; it becomes one measured quantity
Construction null as benchmark	All 91, as a diagnostic	Unchanged	Yes	Changes no correlation; changes the retention rule and how loadings are read

Why consistency fails.

- **The same arithmetic is artifact in one pair and construct in another.** Shared expenses define reserves (`months_cash` $\times$ `liquid_months`) but are noise between `months_cash` and `admin_exp` . A uniform fix removes both or neither.
- **The largest artifacts have no variable-level repair.** Reciprocal and identity links (27 pairs) come from lines sitting inside totals. Removing them means redefining the ratio, not transforming it.
- **`assets_rev` is the ratio of the two scale variables.** Any size adjustment strong enough to clear shared-denominator coupling deletes it.
- **Construction-aware retention is unmoved by every fix.** Against the null, parallel analysis keeps 1 component under F0, F2, and F3 alike.

The consistent element is therefore a benchmark, not a transformation: the construction null applies to every pair in the same way. Transformations then have to be chosen by link type.

Proposal: a construction step before variance factoring

Add one stage between normalization and diagnostics. It audits how the ratios are built, repairs the variable set by link type, and replaces random-data parallel analysis with a construction null. It changes the variables and the retention rule, never individual correlations.

flowchart LR

```
A[990 lines] --> B[Rules and<br/>imputation]
B --> C[Winsorize,<br/>asinh, z]
C --> D[NEW: construction<br/>step]
D --> E[Diagnostics<br/>and triage]
E --> F[Dimension<br/>discovery]
F --> G[Fit, score,<br/>index]
```

The new stage sits where correlation-based decisions start, so triage, retention, and naming all see the benchmark.

1. **Audit links.** Generate the pair table from line-level formulas plus the three identities (A as a sum, $NA = A - L$, $\Delta NA \approx R - E$). Store it with the frame so every later table can cite link type.
2. **Repair the variable set by link type.**
 - *Closure:* replace each closed pair with one bounded contrast, for example (cash – land) / (cash + land). It is defined whenever either part is positive, unlike a log-ratio.
 - *Shared numerator, construct role:* enter the surplus once. Combine `surplus_margin`, `return_assets`, and `na_growth` before factoring, or keep one.
 - *Stock over flow:* decompose each as a share of assets times A/E, so the common scale enters once as its own variable instead of hiding in three ratios.
 - *Reciprocal and identity:* no transformation; carry to step 3. Simulated below, these repairs fix their targets but create composition links, so step 3 carries the weight.

3. **Benchmark against the construction null.** Shuffle size-scaled lines within cells of expenses $\times$ assets per dollar of expenses, at least 20 times, through the full pipeline; this conditional null fit the real data best. Use null eigenvalues for retention. Report each dimension's congruence with its null counterpart. A dimension at 0.85 or above is construction-anchored: its evidence is strength beyond the null, not its loading pattern.
4. **Rule out two shortcuts.** Never factor a pairwise-adjusted matrix. Never residualize on size while `assets_rev` or another scale ratio is in the set.

What this would change in the current model. The four dimensions would not be rejected, but three would be relabeled. Performance is one quantity measured three ways. Asset structure is a closure contrast. Reserves is mainly assets relative to expenses. Leverage is the only dimension whose within-dimension correlations beat the stronger null in every pair, and only by 0.03 to 0.07 (`debt_assets` $\times$ `payables_una` : 0.68 observed, 0.61 null). Two of three performance correlations fall below the null.

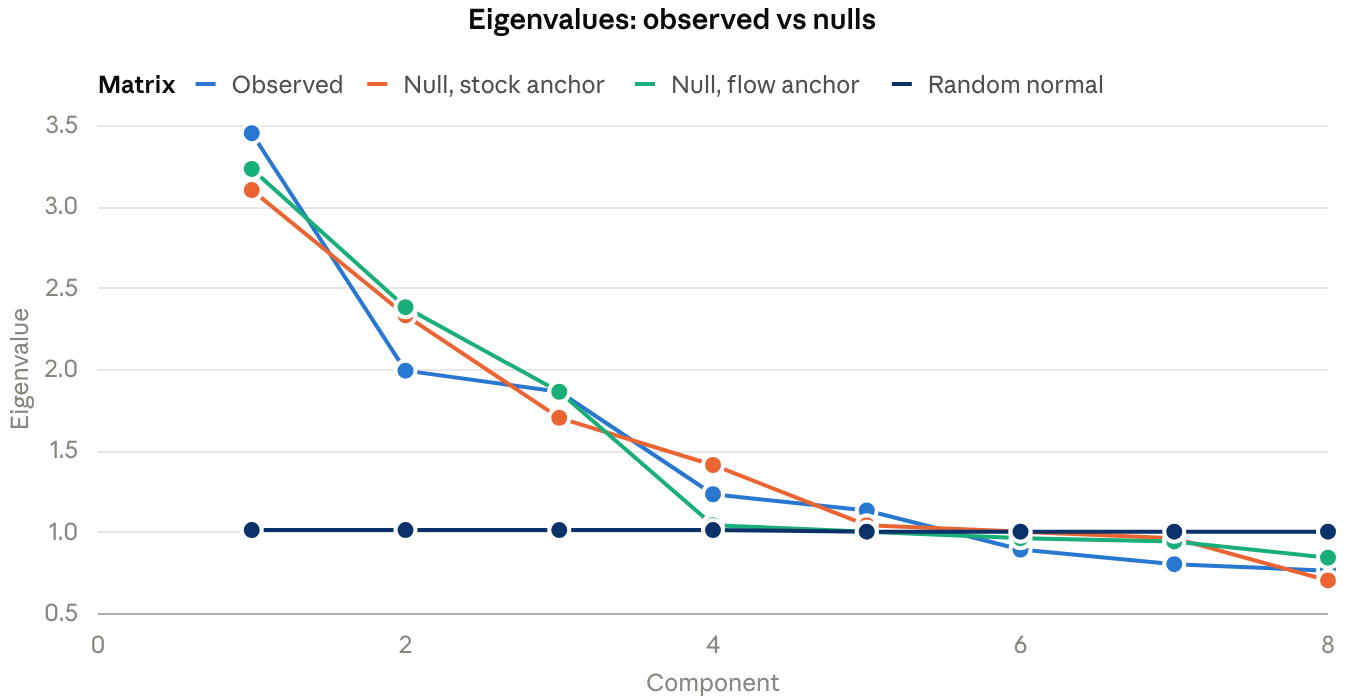
Simulation vignette: shuffled 990 lines reproduce the structure

Shuffling the 990 lines across organizations destroys every real financial relationship. It still reproduces the observed eigenvalues and four of the five loading patterns. The simulation is in `vignettes/ratio-correlation-artifacts.Rmd`, added to the Methods articles in `_pkgdown.yml`.

Design.

- **Data.** Bundled `dat10k` by default (6,207 full-990 filers, 2021). The `efile_path` parameter points it at the deduplicated 2023 file (321,612 filers), which reproduces the five-component table exactly. Figures here are 2023; on `dat10k` the null congruences agree within 0.02 and parallel analysis also keeps 1 component.
- **Pipeline.** 990 lines, then the rules and conditional imputation, then winsorize, asinh, and z-score, copied from `build-ratio-frame.R`. Lines and ratios are separate steps so shuffled lines pass through the identical pipeline.
- **Construction null.** Each line is divided by a size reference, shuffled independently across organizations, and multiplied back. Totals are rebuilt from their parts (A, L, NA, NA at start of year). Two size references: total expenses (flow) and total assets (stock). 20 shuffles for 2023, 30 for `dat10k`.

- **Comparisons.** Null r per pair against the formula prediction; eigenvalues; five-component oblimin loadings with Tucker congruence; fixes F1-F3 applied to both observed and null frames.



2023 full-990 filers; null and random lines are 95th percentiles across shuffles

ratio-correlation-artifacts.Rmd on efile-2023-bmf-dedup.rds · first 8 of 14 components

Findings.

1. **Construction explains the linked pairs.** The null reproduces 95% of observed $|r|$ for closure pairs and more than 100% for reciprocal (108%), identity (112%), and shared-numerator pairs (119%). Above 100% means the data pull those correlations below what arithmetic alone implies. Shared-denominator pairs (38%) and unlinked pairs (21%) carry mostly real covariance.
2. **The null recovers the dimensions.** With assets as the size reference, null loadings match performance at 0.98, reserves 0.90, leverage 0.91, and asset structure 0.95. With expenses as the reference, reserves disappears (0.03). Reserves exists only because expenses vary independently of assets.
3. **Retention collapses to one component.** Standard parallel analysis against random data keeps 5. Against the construction null it keeps 1, under every fix tested.
4. **The formula audit mostly holds.** Predicted signs matched the null for 28 of 32 pairs with $|null\ r| \geq 0.05$, and the null found 4 missed links through una and A.

Render on the 2023 file:

```
rmarkdown::render("vignettes/ratio-correlation-artifacts.Rmd",
  params = list(efile_path = file.path(tools::R_user_dir("fiscal", "cache"),
                                         "efile", "efile-2023-bmf-dedup.rds"),
               reps = 20))
```

Step 2 simulated: the redesign moves the artifact instead of removing it

The redesigned 10-variable set repairs each link it targets, yet its construction correlation is no lower than the original 14 ratios. Putting every balance-sheet quantity over total assets turns stock-over-flow coupling into composition, where closure and identity links are stronger. Results are from section 9 of the vignette on 2023 filers.

The redesign. `cash_assets` and `land_assets` become one contrast, `cash_land` = (cash – land) / (cash + land). `admin_exp` and `fundr_exp` become `admin_fundr` = (M – Fu) / (M + Fu). The three performance ratios become one `surplus` score. `months_cash`, `liquid_months`, and `assets_rev` become two asset shares, `liquid_assets_share` and `una_liquid_share`, plus one scale, `assets_exp` = A / E. `current`, `debt_assets`, `payables_una`, and `self_suff` are unchanged.

Set	Variables	Mean r observed	Mean null r (flow / stock)	Pairs with null r ≥ 0.10	Components kept (random / null flow / null stock)
As built	14	0.175	0.139 / 0.118	37%	5 / 1 / 1
Redesigned	10	0.208	0.153 / 0.121	49%	3 / 1 / 3

Targeted links were repaired.

- **Stock over flow.** The null correlation between `months_cash` and `liquid_months` was 0.50 / 0.68. Between their asset-share counterparts it is 0.16 / 0.22, while the observed correlation is 0.43.
- **Shared numerator and closure.** The three surplus pairs and the –0.51 cash–land pair no longer exist as pairs.

New links replaced them.

- `liquid_assets_share` × `cash_land` : null 0.66 / 0.69, because cash is in both.
- `una_liquid_share` × `debt_assets` : null –0.60 / –0.67, because una = A – L – RES.

- `una_liquid_share` × `payables_una` : null -0.41 / -0.46; `una_liquid_share` × `cash_land` : null 0.40, through land.

Structure. Parallel analysis against random data keeps 3 components. Two are still predicted by arithmetic.

Component	Main observed loadings	Null congruence (flow / stock)
Balance-sheet strength	<code>debt_assets</code> -0.82, <code>payables_una</code> -0.81, <code>current</code> 0.69, <code>una_liquid_share</code> 0.65, <code>assets_exp</code> 0.51, <code>surplus</code> 0.43	0.95 / 0.94
Liquid asset mix	<code>liquid_assets_share</code> 0.89, <code>cash_land</code> 0.78, <code>assets_exp</code> -0.69, <code>una_liquid_share</code> 0.50	0.95 / 0.90
Revenue and expense model	<code>self_suff</code> 0.77, <code>admin_fundr</code> 0.74	0.22 / 0.17

Performance is no longer a dimension once the surplus enters once. Leverage and reserves merge into one balance-sheet component. The only component arithmetic does not predict pairs two variables correlated at 0.22.

What the data add beyond arithmetic. The largest gaps between an observed correlation and the nearer null run across the old dimension boundaries, and none exceeds 0.32.

Pair	r observed	Beyond null
<code>assets_exp</code> × <code>surplus</code>	0.21	+0.32
<code>cash_land</code> × <code>debt_assets</code>	-0.25	-0.30
<code>una_liquid_share</code> × <code>cash_land</code>	0.63	+0.23
<code>self_suff</code> × <code>admin_fundr</code>	0.22	+0.22
<code>liquid_assets_share</code> × <code>debt_assets</code>	-0.14	-0.21

Implications.

- **No variable-level redesign escapes the balance sheet.** $A = L + NA$ makes any set of balance-sheet ratios a composition, so some construction correlation is unavoidable.
- **Construction-predicted components should be reported as indicators, not latent factors.** Their loading pattern is fixed by definitions; only their strength beyond the null is evidence.
- **The real signal is modest and cross-cutting.** Organizations with more assets per dollar of spending run larger surpluses, and cash-heavy asset mixes go with lower debt, once arithmetic is accounted for.

Fixed-construction factor model: works when the null is right, and the real-data null is not

Holding construction correlation fixed inside a factor model is the only method tested that separates planted factors from arithmetic. It depends entirely on the construction null being correct. On real 2023 data neither size reference is correct: fit stays poor and 18 of 23 factors collapse onto a single variable. Results are from section 10 of the vignette.

The model. Let C be the off-diagonal of the construction-null correlation matrix. Fit $\Sigma = \Lambda\Lambda' + \Psi + C$ to the observed matrix by maximum likelihood, with C fixed. Factors only have to explain covariance the arithmetic does not. Unlike pairwise subtraction, the model is fit to a valid matrix.

The planted test. Some 990 lines are reassigned through a Gaussian copula driven by three known factors: liquidity (cash, receivables up; payables down), capital (land, mortgages, depreciation up), and performance (revenue over expenses, other net-asset change up). Every line keeps its exact 2023 distribution. The planting uses the expenses-anchored shuffle, so the flow null is the correct one and the stock null is deliberately wrong. A control scenario plants nothing.

Method	Congruence with planted structure, planted data	Same fit, nothing planted	Detection margin
Ordinary ML factor analysis	0.89	0.86	0.03
Fixed construction, correct (flow) null	0.70	0.29	0.41

Method	Congruence with planted structure, planted data	Same fit, nothing planted	Detection margin
Fixed construction, wrong (stock) null	0.59	0.39	0.20

- **Ordinary factor analysis "recovers" the planted factors from arithmetic alone.** It scores 0.86 on data where nothing was planted, so its 0.89 is not evidence of detection.
- **With the correct null, the fixed-construction model is the only real detector.** With nothing planted, its zero-factor model already fits almost perfectly (SRMR 0.002). With three planted, fit improves from SRMR 0.043 at zero factors to 0.019 at three.
- **With the wrong null, most of that advantage disappears** (margin 0.20), and zero factors fit poorly even when nothing is planted (SRMR 0.102).
- **Eigenvalue-based retention is not reliable at this sample size.** With the correct null it keeps 1 factor when nothing is planted and 6 when 3 are. On `dat10k` it kept exactly 0 and 3; at 321,612 filers, small misfit clears any sampling threshold.

The real data. Three sets were fit: the 14 as-built ratios, the step 2 redesign, and a composition set. The composition set expresses the balance sheet as shares of total assets from both partitions: liquid, fixed, liabilities, unrestricted net assets. Flow measures are relative to expenses.

Set	SRMR, zero factors (flow / stock null)	SRMR, four factors (flow / stock null)	Ordinary FA, four factors	Single-variable factors
As built (14 ratios)	0.135 / 0.143	0.087 / 0.096	0.046	6 of 8
Redesigned (10)	0.139 / 0.130	0.080 / 0.059	0.028	4 of 7
Composition (9)	0.138 / 0.130	0.052 / 0.053	0.021	8 of 8

- **Real data depart from both nulls about three times as much as the planted data depart from the correct one** (SRMR 0.13–0.14 versus 0.043 at zero factors).

- **That departure is not common-factor shaped.** Four factors leave SRMR at 0.05–0.10, and most factors load 0.95 or more on one variable (`debt_assets`, `surplus_margin`, `months_cash`, `liquid_months`, `assets_rev`, `self_suff`, `liab_share`, `una_share`). A single-variable factor absorbs one ratio's misfit to the null; it is not a shared dimension.
- **The composition set does not change this.** Its largest correlation, liabilities share × unrestricted share (–0.82 observed), is close to its null (–0.91). Its beyond-null relationships are the ones found earlier: assets per dollar of spending with surplus (0.29 against a null of 0.00), fixed assets with liabilities (0.22 against –0.09).

Implications.

- **The method is sound; the null is the bottleneck.** A null built on "lines scale with expenses" or "lines scale with assets" does not describe how real balance sheets are built. The observed relationship between expenses and assets, which both nulls break, is the likeliest culprit.
- **Next test: a conditional permutation null.** Shuffle lines only among organizations in the same joint expenses-by-assets cell, so the null keeps the real size structure. Then rerun the planted and real-data fits.
- **Until then, report the fixed-construction model as a diagnostic.** Single-variable factors flag ratios whose construction correlation is mis-modeled, which is useful for triage but not for scoring.

Conditional null: the best benchmark, but still not a scoring model

Shuffling lines only among organizations of similar size and asset intensity gives the best-fitting null so far. It reproduces the four dimensions almost exactly, so their loading patterns are explained by arithmetic plus size structure. The fixed-construction factor model improves under it but still cannot serve as the index's measurement model. What survives beyond this null is a handful of pairwise relationships, not latent dimensions. Results are from section 11 of the vignette on 2023 filers.

The null. Organizations are grouped into 400 cells: 20 bins of total expenses, then 20 bins of assets per dollar of expenses within each (about 800 filers per cell). Each keeps its own expenses. Balance-sheet lines are scaled by assets, flow lines by expenses, and every line is shuffled only within its cell. This null is stricter than the earlier two: it keeps the arithmetic and every relationship that runs through size and asset intensity.

It absorbs the dimensions.

Null	Mean null r (observed 0.175)	Components kept	Congruence: performance / reserves / leverage / asset structure
Flow (expenses)	0.139	1	0.97 / 0.03 / 0.71 / 0.90
Stock (assets)	0.118	1	0.99 / 0.90 / 0.91 / 0.95
Conditional	0.137	1	0.99 / 0.98 / 0.97 / 0.98

Shared-denominator pairs absorb the most: mean null $|r|$ rises from 0.04 to 0.07. For example, `months_cash` × `surplus_margin` (0.36 observed) has a null of 0.00 under both earlier nulls and 0.20 under this one. About half of that link runs through size and asset intensity.

Planted test, rerun. Factors were planted inside the same cells, so the conditional null is the correct one and the global flow null is wrong.

Method	Planted data	Nothing planted	Detection margin
Ordinary ML factor analysis	0.86	0.83	0.03
Fixed construction, conditional null (correct)	0.63	0.43	0.19
Fixed construction, flow null (wrong)	0.46	0.38	0.08

- **The method still separates planted factors from arithmetic, at half the strength.** The margin is 0.19 here against 0.41 when the correct null was the simple flow shuffle.
- **Even the correct conditional null leaves some misfit.** Zero factors fit the nothing-planted data at SRMR 0.026, against 0.002 for the correct flow null in section 10. Cells rebuilt from the data only approximate the cells that generated it.
- **Retention remains unusable.** Eigenvalue rules keep 0 or 5–6 factors whatever was planted.

Real data, rerun.

Set	SRMR, zero factors: conditional null (earlier nulls)	SRMR, four factors: conditional null (ordinary FA)
As built (14 ratios)	0.090 (0.135–0.143)	0.049 (0.046)
Redesigned (10)	0.092 (0.130–0.139)	0.034 (0.028)
Composition (9)	0.092 (0.130–0.138)	0.052 (0.021)

- **Misfit to the null drops by a third**, but zero factors still leave 3.5 times the misfit of the nothing-planted data (0.090 against 0.026).
- **The leftover is still not common-factor shaped.** 9 of 12 factors load 0.90 or more on a single variable: `self_suff`, `land_assets`, `surplus_margin`, `cash_land`, `assets_exp`, `fixed_share`, `fundr_exp`.

What survives the conditional null. These relationships exceed arithmetic plus size and asset intensity, and they recur across the three variable sets.

Relationship	Pair (set)	r observed	Conditional null	Beyond null
Fixed assets financed by debt	<code>debt_assets</code> × <code>land_assets</code> (as built)	0.22	−0.05	+0.28
Fixed assets financed by debt	<code>fixed_share</code> × <code>liab_share</code> (composition)	0.22	−0.04	+0.26
Earned revenue, less fundraising	<code>self_suff</code> × <code>fundr_exp</code> (as built)	−0.21	0.03	−0.24
Cash-rich, less debt	<code>cash_assets</code> × <code>debt_assets</code> (as built)	−0.23	−0.03	−0.21
Liquid assets held as unrestricted equity	<code>liquid_share</code> × <code>una_share</code> (composition)	0.18	0.00	+0.17
Surpluses build cash	<code>months_cash</code> × <code>surplus_margin</code> (as built)	0.36	0.20	+0.17

Implications.

- **Use the conditional null as the benchmark in step 3.** It fits the real data best and is the most conservative.
- **The four dimensions are construction plus size structure.** Their loading patterns match the conditional null at 0.97 or above. They are defensible as indicator groups, but not as evidence of latent constructs.
- **The substantive signal is asset-liability matching and revenue model.** Fixed assets go with debt, liquid assets with unrestricted equity, and earned revenue with less fundraising. A health index built on evidence would test these relationships directly.
- **The fixed-construction factor model stays a diagnostic.** Its detection margin halves under a realistic null, retention rules fail, and most factors are single-variable.

Pre-specification: testing the surviving relationships directly

Five hypotheses replace the factor interpretation, each written in dollar lines so no line sits on both sides of an equation. This section was fixed on 2026-09-17, before any of these models were estimated. H1-H3 are run first; H4 and H5 follow.

Lines (Part X unless noted). Long-term debt = tax-exempt bonds + secured mortgages and notes + unsecured notes (lines 20, 23, 24). Short-term liabilities = accounts payable + grants payable + deferred revenue (17-19). Cash = cash + savings (1-2). Non-cash current assets = pledges + accounts receivable + inventories + prepaid expenses (3, 4, 8, 9). Fixed assets = land, buildings, and equipment, net (10c). Other assets = total assets minus those four, floored at 0.

Common specification.

- **Sample:** 2023 full-990 filers with expenses and total assets above zero.
- **Two parts per outcome:** a linear probability model for any positive amount, and log amount among filers with a positive amount.
- **Predictors:** $\log(1 + x)$ plus an indicator for $x > 0$, for fixed assets, cash, and non-cash current assets.
- **Size controls:** 4-df splines of log expenses and $\log(1 + \text{other assets})$, plus an indicator for zero other assets. Total assets is excluded because it contains the predictors.
- **Evidence rule:** a coefficient counts only if it is in the predicted direction and beyond the 0.5th or 99.5th percentile of 200 refits. Each refit shuffles the predictor within the 400 expenses $\times$ asset-intensity cells. Robust standard errors are reported alongside.

- **Replication:** 2021 `dat10k`, which lacks bonds, unsecured notes, and deferred revenue. Long-term debt there is mortgages only; short-term is payables plus grants payable. Replication succeeds if the sign matches and the 2.5th/97.5th permutation bound is passed.

Hypothesis	Outcome	Coefficient that carries the test	Support if	Reject if
H1 Long-term matching	Long-term debt: any, and log amount	log fixed assets	> 0 in both parts; elasticity in log amount larger for long-term debt than for short-term liabilities	≤ 0 , or no larger than in the short-term equation
H1a Subsector	Long-term debt, log amount	log fixed assets $\times$ capital-intensive subsector (hospitals, universities, housing)	> 0	≤ 0
H2 Short-term matching	Short-term liabilities: any, and log amount	log non-cash current assets	> 0 in both parts; larger than in the long-term equation	≤ 0 , or no larger than in the long-term equation
H3 Pecking order	Long-term debt: any, and log amount	log cash	< 0 in both parts	> 0 in both parts supports the rival collateral account; mixed signs are inconclusive
H4 Revenue model	Fundraising expense: any, and log amount	log program service revenue, holding log contributions and size	< 0; contributions coefficient > 0	≥ 0

Hypothesis	Outcome	Coefficient that carries the test	Support if	Reject if
H5 Surplus allocation	Change from start to end of year in cash, other current assets, fixed assets, and debt	Revenue minus expenses	Shares sum to about 1; the cash share is the estimate of interest	No prediction on the split

Descriptive, not decisive: an H1 elasticity between 0.5 and 1.5 is read as "near 1".

Interpretation: all of these are associations. Borrowing and buying property are decided together, and the choice to own rather than lease inflates H1.

Results: asset-liability matching holds; pecking order is half supported

Both matching hypotheses are supported in 2023 and replicate in 2021. Long-term debt tracks fixed assets and short-term liabilities track non-cash current assets, each far more strongly than in the other equation. Pecking order is inconclusive, and the subsector prediction is rejected. The analysis is in [dev/health-index/matching-hypotheses-2023](#). Rmd : 316,837 filers in 2023 and 6,131 in 2021.

Hypothesis	Test	2023 estimate	Shuffle baseline (median)	2023 verdict	2021 estimate	Replicates
H1 Long-term matching	Any long-term debt, per log unit of fixed assets	+0.066	+0.019	Supported	+0.062	Yes
	Long-term debt amount, elasticity	0.51	0.15		0.54	Yes
	Long-term minus short-term elasticity	+0.45	+0.13		+0.50	Yes
H2 Short-term matching	Any short-term liabilities, per log unit of non-cash current assets	+0.021	+0.002	Supported	+0.023	Yes

Hypothesis	Test	2023 estimate	Shuffle baseline (median)	2023 verdict	2021 estimate	Replicates
	Short-term liabilities amount, elasticity	0.20	0.01		0.19	Yes
	Short-term minus long-term elasticity	+0.20	+0.00		+0.21	Yes
H3 Pecking order	Any long-term debt, per log unit of cash	-0.026	-0.004	Inconclusive	-0.014	Yes
	Long-term debt amount, elasticity	+0.002	+0.041		-0.008	No
H1a Subsector	Elasticity difference, capital-intensive subsectors	-0.13	-0.13	Rejected	-0.07	No

Reading the results.

- **Matching is real, not arithmetic or size.** A 1% difference in fixed assets goes with 0.51% more long-term debt among borrowers, against 0.15% from size structure alone. Short-term liabilities barely respond to fixed assets, and long-term debt barely responds to receivables and inventory. The cross-prediction the theory requires holds in both years.
- **Debt grows less than proportionally with property.** An elasticity of 0.51 sits at the bottom edge of the pre-specified "near 1" range. Larger property holders finance relatively more of it with net assets.
- **Cash lowers the chance of borrowing, not the amount.** Cash-rich filers are less likely to carry any long-term debt. Among borrowers the cash coefficient is about zero, so the pre-specified verdict is inconclusive. Post hoc: that zero is below the size baseline of +0.04, which leans toward pecking order, but it was not the pre-specified test.
- **Matching is not stronger in hospitals, universities, and housing.** The negative interaction equals its shuffle baseline, so the subsector difference is size structure, not a distinct financing pattern.

Notes on the rule. The shuffle distributions are centered away from zero. Shuffling within a cell keeps each line's link to the cell's total assets, which is not a control, so the estimate must have the predicted sign and clear the baseline. A first draft of the code checked only the baseline and briefly marked H3 as supported on a wrong-signed estimate; the verdicts above use the pre-specified rule.

What this gives the index. Leverage can be measured against a confirmed benchmark instead of a factor. The H1 and H2 models predict how much long-term and short-term debt an organization's asset mix supports. The candidate indicator is the gap: short-term liabilities funding fixed assets, or long-term debt beyond what property supports.

Matching-gap indicator: construction and validation plan

The indicator measures how much debt an organization carries beyond what its asset mix and size support. This plan was written before the indicator was computed or validated.

Construction.

1. **Benchmark models.** Refit the H1 and H2 two-part models on 2023 end-of-year lines, with the same predictors and size controls. The any-debt part is logistic, so predictions stay between 0 and 1. The amount part is OLS on log debt, retransformed with Duan's smearing factor.
2. **Predicted debt.** Expected long-term debt = $P(\text{any}) \times E(\text{amount} \mid \text{any})$, and the same for short-term liabilities.
3. **Gaps, as shares of total assets.**
 - `gap_long` = (actual – predicted long-term debt) / total assets.
 - `gap_short` = (actual – predicted short-term liabilities) / total assets.
 - `gap_total` = their sum.
 - Positive values mean more debt than the asset mix supports.

Validation, one year ahead within the same filing. Gaps are computed from beginning-of-year Part X lines, using the end-of-year coefficients. Three outcomes are measured over the following year:

- an operating deficit (revenue below expenses);
- a decline in unrestricted net assets;
- cash falling by more than one month of expenses.

Expected before looking. Each gap predicts a higher probability of each outcome, net of beginning-of-year `debt_assets` and size. `gap_short` is expected to predict the cash drawdown most strongly, because payables beyond working assets must be paid from cash. Support requires a positive coefficient that clears robust 99% intervals, plus a gain in AUC over `debt_assets` alone. A gap with no incremental value would mean the benchmark only relabels leverage.

Matching-gap results: not ready for the index

The matching gap adds almost nothing beyond `debt_assets`. The one planned result that looked strong, cash drawdowns, was mechanical. The long-term gap carries a small, robust signal about deficits and net-asset declines. The short-term gap points the wrong way. The analysis is in `dev/health-index/matching-gap-2023.Rmd`: 316,837 filers for the benchmarks and 311,865 for validation. The indicator is saved to the fiscal cache as `matching-gap-2023.rds`.

Planned results. Two-part benchmark, planned controls; outcome base rates are 40% deficit, 37% net-asset decline, 25% cash drawdown.

Outcome	gap_long per SD	gap_short per SD	AUC, debt_assets alone	AUC gain from gaps	Met the plan
Deficit	+0.008	-0.003	0.548	-0.000	gap_long only; no AUC gain
Net-asset decline	+0.011	-0.007	0.586	+0.001	gap_long only; no AUC gain
Cash drawdown	+0.037	+0.037	0.574	+0.041	Yes, and gap_short strongest here, as expected

Two problems found after the first run, with added checks.

- **The planned benchmark is badly calibrated.** Predicted totals are 2.3–2.4 times actual, because log residuals are so dispersed that the smearing factor is about 4. A Poisson pseudo-likelihood (PPML) benchmark matches totals by construction and tracks the top decile, but still over-predicts the middle deciles.
- **The cash drawdown outcome has a floor.** Cash cannot fall by a month of expenses unless the organization starts with that much. Negative gaps mark organizations with few assets, and so little cash, relative to expenses.

Benchmark and controls	Cash drawdown: gap_total per SD (AUC gain)	Deficit: gap_long per SD	Decline: gap_long per SD	gap_short for deficit / decline
Two-part, planned	+0.057 (+0.041)	+0.008	+0.011	-0.003 / -0.007

Benchmark and controls	Cash drawdown: gap_total per SD (AUC gain)	Deficit: gap_long per SD	Decline: gap_long per SD	gap_short for deficit / decline
Two-part, + cash months and asset intensity	+0.001 (0.000)	+0.012	+0.010	-0.007 / -0.012
PPML, planned	+0.040 (+0.019)	+0.028	+0.035	-0.015 / -0.021
PPML, + cash months and asset intensity	-0.003 (0.000)	+0.021	+0.029	-0.013 / -0.022

Reading the results.

- **The cash drawdown result was the floor.** Controlling for starting cash and asset intensity removes it entirely under both benchmarks. It was the only outcome with a meaningful AUC gain.
- **The long-term gap is real but small.** Borrowing beyond what property supports raises next-year deficit and decline probabilities by 2–3 points per SD under PPML, about 5–8% of the base rate. It clears 99% intervals everywhere, but no row adds more than 0.0033 AUC.
- **The short-term gap measures activity, not strain.** More short-term liabilities than current assets support goes with fewer deficits. Deferred revenue and payables grow with operations, so they mark busy organizations, not stretched ones.
- **Scaling by total assets distorts small-asset filers.** Predictions follow expenses, so organizations with few assets get extreme negative gaps (PPML 1st percentile: -14 times assets).

Recommendation. Keep `debt_assets` as the leverage measure; don't add `gap_total` or `gap_short` to the index. If the long-term gap is pursued, four changes come first:

- use the PPML benchmark;
- scale by expenses or by predicted debt, not by assets;
- separate accounts payable from deferred revenue;
- validate against multi-year outcomes such as net-asset erosion or stopping filing, from a panel.

H5 results: a surplus dollar becomes 37 cents of cash

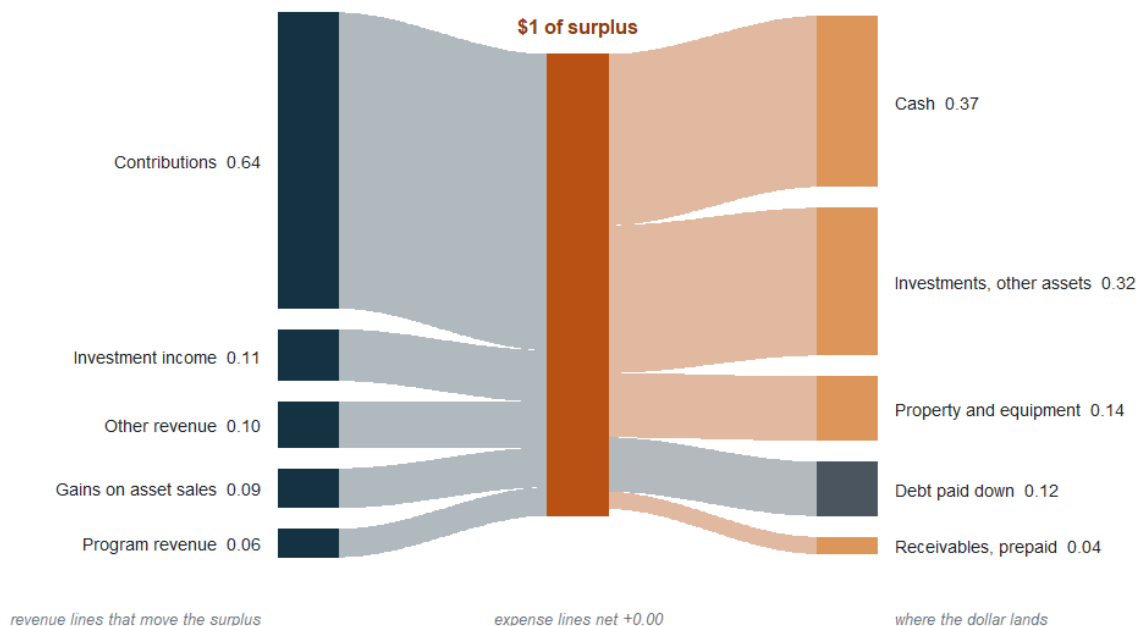
Surpluses go mostly to cash and investments, barely to debt. Deficits are absorbed by cash at nearly twice the rate. The accounting identity holds exactly in the estimates, which is what makes this the most self-checking test of the five. The analysis is in `dev/health-index/h5-surplus-allocation.Rmd`, on the 2019–2024 panel: pooled tax years 2022–2023, December year-ends, organizations filing a full 990 every year 2019–2023. That is 247,124 organization-years from about 127,000 organizations, after dropping 2.9% of rows as outliers.

Where the dollar lands	Pooled 2022–23	Within-organization	2019	2021 (relief year)
Cash	0.371	0.434	0.342	0.390
Investments and other assets	0.320	0.321	0.525	0.495
Property	0.139	0.154	0.214	0.100
Receivables, inventory, prepaid	0.037	0.046	0.024	0.033
Long-term debt paid down	0.021	0.018	0.012	0.011
Short-term liabilities paid down	0.010	0.011	0.002	0.000
Other liabilities paid down	0.087	0.081	0.008	0.013
Identity: assets – liabilities	0.985		1.126	1.042

Where it comes from. The same decomposition run on the inflow side splits a dollar of surplus variation across the revenue and expense lines, and those shares add to exactly 1. Contributions account for 0.64, investment income 0.11, other revenue 0.10, gains on asset sales 0.09, and program revenue 0.06. The four expense lines net to 0.00. Surpluses are revenue events: organizations with unusually good years are not the ones that spent unusually little.

Where a surplus comes from, and where it goes

cents per dollar of surplus | full-990 filers, December year-ends, 2022-2023 | n = 247,124



What it says.

- **Liquidity first, debt last.** About 37 cents of each surplus dollar ends the year as cash and 32 cents as investments. Only about 3 cents goes to paying down identifiable debt (the 8.7 cents on other liabilities has a standard error of 0.069 and is not distinguishable from zero).
- **The identity holds exactly.** Asset coefficients minus liability coefficients equal the direct net-assets regression to three decimals (0.985), and the balance-sheet blocks add up in essentially every filing (100% for assets and liabilities, 99.9% for net assets).
- **Year-to-year variation is the investment market, not behavior.** The investments coefficient is 0.165 in 2022 with an identity sum of 0.774, against 0.479 in 2023 with 1.202. Net assets fell short of the surplus in the down market and exceeded it in the rebound.
- **Deficits are absorbed by cash, not borrowing.** A dollar of deficit draws 0.575 from cash, 0.273 from investments, and 0.089 from property, while debt does not move (+0.005). Organizations spend down liquid assets rather than borrow, which matches the H3 finding that cash-rich filers are less likely to carry debt.
- **Within-organization estimates agree.** Restricting to variation inside one organization across 2019, 2022, and 2023 gives 0.434 to cash and 0.154 to property, close to the pooled split.

Two method notes.

- **Winsorizing broke the identity.** A first version clipped each component at the 1st and 99th percentiles. Clipped parts no longer sum to the clipped total, and the identity check failed immediately (0.595 against 0.905). Dropping whole rows instead keeps the components additive, and the check then passes exactly.
- **Trimming is not cosmetic.** Untrimmed, the estimate is 0.97 to cash and about zero everywhere else: a handful of organizations with enormous surplus-to-expense ratios dominate. The identity holds either way, which is a reminder that it validates internal consistency, not representativeness.

For the index. The "surpluses build cash" link, +0.17 beyond the conditional null, is now a magnitude: 37 cents on the dollar. The asymmetry suggests a sharper distress indicator than either piece alone, since an organization running deficits with little cash has no other margin. Debt is not the shock absorber.

How this relates to the four dimensions. H5 does not test the performance dimension. It tests the *transmission* from that dimension to the other three, and that reframes what the dimensions are.

- **Performance is the only flow among them.** Reserves, leverage, and asset structure are all stocks measured at a moment. H5 says what the flow becomes: 37 cents of cash, 32 of investments, 14 of property, 12 of debt repayment.
- **Reserves are accumulated performance.** About 69 cents of every surplus dollar lands in liquid or investment assets, so the reserves dimension is largely the running total of past surpluses plus market returns. That is why `months_cash` × `surplus_margin` survives the conditional null at +0.17. The two dimensions are not independent evidence about an organization, and an index that weights them separately counts the same history twice.
- **Leverage is genuinely separate, and now for a behavioral reason.** Only about 3 cents of a surplus touches identifiable debt, and deficits are absorbed by cash rather than borrowing. Leverage stands apart from performance in the factor solution because organizations do not use surpluses to manage debt, not only because of how the ratios are built.
- **Slack is a stock, not a flow.** A single year's surplus is not slack; slack is what the surpluses accumulated into. That argues for measuring the reserves dimension as the health signal and treating surplus margin as the annual increment to it, rather than as a separate dimension of equal standing.

H4 results: a business model, not a behavior

The pre-specified test is supported: organizations that earn more of their revenue spend less on fundraising. The panel then shows the relationship is between organizations, not within them. When one organization's earned revenue moves, its fundraising moves the same way, not the opposite way, and organizations that lost earned revenue in the pandemic did not raise fundraising spending afterwards. The analysis is in `dev/health-index/h4-revenue-model.Rmd`.

Test A, pre-specified (2022–2023, December year-ends, full-990 stable filers).

Outcome	log program revenue	Shuffle bound	log contributions	Verdict
Any fundraising expense (n = 253,671)	−0.0366	−0.0020	+0.0693	Clears
Log fundraising expense (n = 84,588)	−0.0622	−0.0067	+0.2800	Clears

Both parts have the predicted sign and clear the within-cell shuffle bound by a wide margin, and contributions predict fundraising positively as expected. **H4 is supported.**

Within organizations, the sign flips. Removing each organization's mean across 2019, 2022, and 2023, a one-unit rise in log program revenue goes with a **+0.014** change in log fundraising expense (SE 0.002), not a fall. Contributions behave the same way (+0.025). Earned revenue and fundraising rise together inside an organization; they trade off only across organizations.

The COVID shock (exploratory). Among 79,895 organizations with earned revenue in 2019, the change in program revenue from 2019 to 2021 does not predict a rise in fundraising spending by 2023 (+0.005 per unit, SE 0.005). Two secondary results:

- Organizations that lost more earned revenue received **more** contributions by 2023 (−0.066 per unit, SE 0.006, so a standard-deviation loss goes with a 0.15 rise in log contributions).
- Among the 54,638 reporting no fundraising expense in 2019, a larger loss slightly raised the chance of starting (−0.003 per unit, SE 0.0006).

Read the contributions result carefully. Relief funding is the obvious rival explanation: organizations whose earned revenue collapsed were also the likeliest to receive PPP and emergency grants, much of which was recognized as contributions. The design has no

pre-period trend and subsector fixed effects only partly absorb differential exposure, so this is suggestive at best.

What it means for the index. `self_suff` and `fundr_exp` measure a revenue-model type, not a management decision. They describe what kind of organization this is, stably over years, and they do not indicate that a nonprofit under pressure is pulling a lever. That fits their position in the factor solution, where they formed the one component arithmetic could not predict, and it argues for reporting them as context alongside the health dimensions rather than inside a directional score.

Matching gap on the panel: a posture, not a warning

Measured in 2019 and used to predict 2020–2023, the long-term matching gap behaves like a standing financing posture. It is highly persistent, its risk signal runs in both directions, and it adds essentially nothing to a model that already knows leverage, size, cash, and asset intensity. The analysis is in `dev/health-index/matching-gap-panel.Rmd`: 126,746 organizations with a full 990 and a December year-end in every year 2019–2023.

Base rates over 2020–2023. 20.8% ran three or more deficit years out of four, 28.6% ended 2023 with lower net assets than 2019, and 12.7% had not filed for 2024 when the file was built (which overstates exit, since 2024 is incomplete).

The gap is a trait. Spearman correlation between 2019 and 2023: 0.71 for the expense-scaled gap, 0.64 for the asset-scaled version, against 0.78 for `debt_assets` itself. An organization's position relative to the benchmark barely moves across four years and a pandemic.

Outcome, 2020–2023	Rate	Expense-scaled gap, per SD	AUC without the gap	AUC gain
Three or more deficit years	0.208	+0.020 (SE 0.002)	0.600	+0.0001
Two deficits in a row	0.344	+0.016 (SE 0.002)	0.570	+0.0002
Net assets fell	0.286	+0.017 (SE 0.002)	0.564	+0.0005
Cash fell by a month of expenses	0.228	−0.002 (SE 0.002)	0.739	+0.0000
No 2024 filing	0.127	+0.001 (SE 0.001)	0.540	+0.0001

Scaling by expenses is the better choice, as the memo suggested. The asset-scaled gap is flat or wrong-signed for every outcome; the expense-scaled version at least points the right way.

Both tails carry risk, which a linear term hides. Within the high-leverage third, organizations in the top gap quintile ran three-plus deficits 32.4% of the time against 16.6% in the fourth quintile, while in the middle third the *lowest* gap quintile was worst at 27.1%. Entering the absolute deviation from the median gives +0.025 per SD for three-plus deficits (SE 0.002) and +0.029 for net-asset declines (SE 0.002). Being far from the benchmark in either direction is worse than sitting on it: over-borrowed on one side, a thin balance sheet relative to spending on the other.

It still does not earn a place in the index. Even with quintile dummies or the absolute deviation, the AUC gain never exceeds 0.003. The clearest effect is on four-year net-asset growth: a standard deviation of expense-scaled gap goes with -0.069 standard deviations of growth (SE 0.010).

Caveat on the window. The gap is measured in 2019 and the outcomes span the pandemic, so this is a test of whether pre-COVID over-borrowing predicted pandemic-era strain. Relief funding cushioned exactly the organizations that would otherwise have been exposed, which may be part of why the signal is so weak.

H1-H3 on the panel: matching is behavior, but a small part of financing

Matching survives the move inside organizations: when an organization buys property, its long-term debt rises, and the cross-prediction holds in every window. The magnitude is far smaller than the cross-section implied, which changes what the relationship means. The analysis is in `dev/health-index/h1-h3-panel.Rmd`, on the 126,000 filers with a full 990 and December year-end in all of 2019–2023.

Year by year, the cross-sections are stable.

Year	H1: any debt, per log fixed assets	H1: debt amount elasticity	H2: short-term elasticity	H3: any debt, per log cash
2019	+0.071	0.582	0.191	-0.033
2020	+0.059	0.418	0.202	-0.028
2021	+0.064	0.462	0.211	-0.030

Year	H1: any debt, per log fixed assets	H1: debt amount elasticity	H2: short-term elasticity	H3: any debt, per log cash
2022	+0.065	0.482	0.212	-0.031
2023	+0.065	0.478	0.213	-0.031

The 2020 dip in the H1 elasticity is the PPP effect: unsecured notes appear for organizations with no property to match. Cross-checks stay small in every year (long-term debt on current assets 0.004–0.020; short-term liabilities on fixed assets 0.051–0.058), so the pattern the theory requires holds throughout.

Inside organizations, the cross-prediction holds and the horizon matters. Changes are scaled by starting expenses, so each entry is cents of new debt per dollar of new assets.

	1 year (2022–23)	2 years (2021–23)	4 years (2019–23)
New long-term debt per dollar of new property	0.074	0.107	0.111
New long-term debt per dollar of new current assets	0.020	0.036	0.034
New short-term liabilities per dollar of new current assets	0.101	0.107	0.134
New short-term liabilities per dollar of new property	0.012	0.008	0.005

Every coefficient clears a placebo in which asset growth is shuffled among organizations of similar starting size and asset intensity (bounds of roughly ± 0.004). Longer windows give larger coefficients, which is what lumpy purchases and amortizing debt predict: a single year catches only the timing.

The size of the effect is the finding. Only 7 to 11 cents of each new dollar of property arrives as new long-term debt, against a cross-sectional elasticity of about 0.48. Most property growth is funded from net assets, which squares with H5: a surplus dollar puts 14 cents into property. The cross-sectional relationship is mostly organizations sorting into capital-intensive types, not organizations financing purchases with debt.

Dynamic H3 gets weak support at longer horizons. Per standard deviation of starting cash (about 18–24 months), long-term debt grows by -0.0001 of starting expenses over one year, -0.0016 over two, and -0.0057 over four (SE 0.0009). The sign is right and strengthens with the horizon, but the magnitude is trivial: half a percent of one year's expenses. Cash-rich organizations borrow slightly less later; they do not use cash to avoid debt in any material way.

One sample caveat. In this December-only, five-year-survivor sample the H3 debt-amount coefficient is negative from 2020 onward (-0.015 to -0.027), where the 2023 all-filer cross-section gave $+0.002$. The pecking-order reading is sensitive to which filers are in the sample, which is a reason to keep treating H3 as unresolved.

Dimension weighting: keep both, but not equal weights

The double-counting worry does not survive contact with the data. Reserves and performance predict different failures, so both belong in the index. The equal-weight default is still wrong, for a different reason: leverage adds essentially nothing. The analysis is in `dev/health-index/dimension-weighting.Rmd`, scoring the retained indicator sets on the panel and using 2021 scores to predict 2022–2023.

Reserves is a stock; performance is a flow; they overlap less than expected. The two scores correlate at 0.24 (Spearman, 2023). Reserves persists across four years at 0.76, performance at 0.16.

Accumulation is real but slow. The 2019 reserves score explains 59% of the variance in the 2023 score on its own; adding four years of cumulative surplus takes that to 61%. In natural units, a cumulative surplus equal to one year of expenses ends up as 2.2 more months of cash. Over a four-year window the stock is dominated by where it started, not by the surpluses since — so the H5 result about marginal dollars does not make the reserves score a restatement of past performance.

The two dimensions predict different things.

Outcome, 2022–23	Rate	AUC: reserves	AUC: performance	Incremental AUC of performance	Incremental AUC of reserves
Deficit in both years	0.239	0.526	0.621	+0.078	+0.001
Deficit in either year	0.602	0.577	0.603	+0.037	+0.015
Net assets fell	0.442	0.519	0.578	+0.057	−0.002

Outcome, 2022-23	Rate	AUC: reserves	AUC: performance	Incremental AUC of performance	Incremental AUC of reserves
Cash fell by a month	0.226	0.633	0.531	+0.000	+0.098
No 2024 filing	0.127	0.539	0.527	-0.000	+0.010

Performance carries the income-statement outcomes; reserves carries the cash outcome. Leverage adds nothing anywhere (average incremental AUC -0.002), which matches the matching-gap result.

Performance is not a restatement of the change in reserves. Adding the 2019-2021 change in the reserves score to a model that has the reserves level adds no AUC, while adding performance moves the both-year-deficit AUC from 0.527 to 0.623. The annual flow carries information that the two-year change in the stock does not.

Implied weights. Setting weights proportional to average incremental AUC gives performance 0.59, reserves 0.41, and leverage 0.00, against 0.33 each under the default.

Three cautions before adopting those numbers.

- **The cash outcome has a floor.** An organization can only lose a month of cash if it had one, and the reserves coefficient there is positive (higher reserves, more likely to lose a month). That outcome inflates the reserves share; excluding it tilts the weights further toward performance.
- **Performance predicting deficits is partly self-prediction.** Margins are autocorrelated, so a performance score forecasting future deficits is closer to persistence than to insight. Legitimate for early warning, but it should not be read as validation of the construct.
- **The 2024 filing outcome is contaminated** by the incomplete file.

Recommendation. Keep reserves and performance as separate, differently weighted dimensions: roughly 55-60% performance and 40-45% reserves on this evidence. Report leverage and asset structure as context rather than scoring them, since neither earns predictive weight. Do not substitute a change-in-reserves term for performance. And describe the index for what it is: a stock, a flow that feeds it, and two descriptive dimensions that do not move the score.

Reweight index: two thirds of filers move, and equal weights dilute the signal

Rebuilding the index under the implied weights moves a lot of organizations and improves discrimination modestly. The more useful result is a comparison the exercise forced: **the equal-weighted composite predicts operating distress worse than the performance dimension on its own.** The rebuild is in `dev/health-index/index-reweighted.Rmd`, with scoring factored into `dimension-scores.R` so it cannot drift from the weighting analysis.

Three schemes, all over the same dimension scores: equal thirds; incremental AUC (reserves 0.41, performance 0.59, leverage 0); and a sensitivity that drops the floor-affected cash outcome from the weighting (reserves 0.12, performance 0.88).

Rankings move more than the rank correlation suggests. Equal and incremental weights correlate at 0.88 (Spearman), but 64% of organizations change at least one decile, 27% change two or more, and 8% change three or more. The mean move is 10.4 percentile points.

The movement has a clear direction.

	Reserves	Performance	Leverage (higher = less debt)	Median margin	Median cash months
Biggest risers (top 1%)	-0.69	+1.45	-2.34	+0.15	4.0
Biggest fallers (bottom 1%)	+0.90	-1.46	+0.81	-0.78	10.6

Reweighting promotes indebted but profitable organizations and demotes cash-rich ones that are losing money. Whether that is right depends on what the index is for: the fallers are running a median margin of -78% while sitting on ten months of cash, so they are not in immediate danger, but they are not healthy either.

Discrimination improves, modestly. Using the 2021 index to predict 2022-2023:

Outcome	Equal	Incremental	Performance-dominant
Deficit in both years	0.565	0.595	0.622
Deficit in either year	0.566	0.599	0.607
Net assets fell	0.516	0.547	0.574

Outcome	Equal	Incremental	Performance-dominant
Cash fell by a month	0.443	0.431	0.473
No 2024 filing	0.516	0.515	0.511

Three things to take from that table.

- **Equal weights dilute.** The performance dimension alone scores 0.621 on both-year deficits; the equal-weighted composite scores 0.565. Averaging in two dimensions that do not predict costs more than it adds.
- **The performance-dominant composite is no better than performance alone** (0.622 against 0.621). For predicting operating distress, the composite earns nothing over a single dimension.
- **Every version is anti-predictive for cash drawdowns** (AUC below 0.5), because healthier organizations are the ones with a month of cash to lose. That outcome should be dropped from validation work, not just down-weighted.

Triage barely changes. The share of both-year-deficit organizations landing in the bottom decile is 14.9% under equal weights, 14.7% under incremental, and 15.8% under performance-dominant. The deficit rate inside the bottom decile runs 35–38% against 24% overall.

What this settles. If the index is meant to rank fiscal health descriptively, keep the dimensions and use the incremental weights, which at least stop the non-predictive dimensions from diluting the score. If it is meant to flag organizations heading for trouble, no weighting of this composite beats reporting the performance dimension directly, and the honest framing is a small dashboard — performance as the forward-looking measure, reserves as the buffer, leverage and asset structure as context — rather than one number.

Failure test: the index works best on the outcome it was never validated against

Every earlier validation used a panel that required a filing in all of 2019–2023, which removed the failures. A second panel keeps them: all 280,158 organizations that filed a full 990 for 2019, followed through 2024 whether or not they kept filing (`dev/panel/panel_build_exits.R`, built from the cached CSVs with no re-downloading). The analysis is in `dev/health-index/exit-validation.Rmd`.

The outcome. A two-year filing gap covering 2022 and 2023, confirmed by absence in 2024: 5.4% of the cohort. It is not the same as failure. Organizations also disappear by merging, or by shrinking below the 990 threshold onto the 990-N postcard, which is not in these files. Gap rates fall steeply with size, from 12.4% under \$100k to 2.0% at \$10m or more, which is what that confound looks like.

Dimensions predict gaps better than they predict deficits, and the ranking among them flips.

Predictors (all with size splines)	AUC, confirmed gap
Size only	0.662
Size + leverage	0.664
Size + performance	0.689
Size + reserves	0.708
Size + all three	0.715

Reserves is the strongest dimension here, where performance was strongest for deficits. Both coefficients run the right way (higher score, lower gap risk). Leverage runs slightly the wrong way: debt-free organizations disappear a little more often, which fits the picture of small, thinly capitalized nonprofits winding down.

For large organizations the index earns its keep. Among the 91,553 filers with \$1m or more of expenses, where dropping to a postcard is implausible and a gap is closer to real failure:

	AUC
Size only	0.554
Three dimensions	0.639
Index, incremental weights	0.642

That is the first outcome in this whole exercise where the composite clearly beats both size and its own components elsewhere.

Triage improves too. The confirmed-gap rate in the bottom decile is 10.8% under equal weights, 15.7% under incremental, and 16.5% under performance-dominant, against 5.4% overall. Equal weighting again loses to both alternatives.

One oddity worth keeping. Gap rates are U-shaped across the index: the top decile runs 6.9–7.8%, above the middle deciles. The healthiest-looking small organizations — cash-rich, debt-free, tiny — also disappear at elevated rates. A single directional score cannot represent that, which is another argument for reporting size and scale alongside any index rank.

Caveat on the 2024 file. It is incomplete, so some organizations counted as confirmed gaps may yet file. Since a confirmed gap already requires two prior missing years, this mostly adds noise rather than bias.

Limits, open questions, references

The construction null is a model, not a proof. Three of its assumptions shape the findings, and the proposed repairs are untested.

Limits.

- **Flow lines are drawn relative to expenses in both nulls.** That matches the accounting (functional expenses sum to E), but it sets shared-E coupling among `admin_exp`, `fundr_exp`, `self_suff`, and `surplus_margin` to zero by assumption.
- **The null keeps each line's dispersion.** Coupling strength depends on that dispersion, so the null is only as realistic as the claim that lines vary independently given size.
- **Structured nulls push late eigenvalues below 1.** Retention compares only the leading run of eigenvalues. Revised parallel analysis, which conditions on earlier factors, would be the formal version.
- **Imputation injects correlation in both frames.** `cond_impute` predicts missing values from `liquid_months`, `surplus_margin`, `months_cash`, and `self_suff`. That link is present in observed and null data alike, but it is not separated out.
- **Shared ingredients are not meaningless.** The surplus is a real quantity. The finding is that three rescalings of it are not independent evidence of a dimension.

Open questions.

- ☐ Report both size references, or pick one as the index's official null?

- ☐ Step 2, the balance-sheet composition set, and the fixed-construction factor model are tested above. The conditional null is tested too. H1-H3 are tested (above). The matching gap is built and validated (above). H5 is tested on the 2019-2024 panel (above). H4 is tested on the panel (above). The gap is validated on the panel (above). H1-H3 are rerun on the panel (above). Dimension weighting is settled above. The index is rebuilt and compared (above). The filing-gap test is done (above). Next: separate true exits from 990-N downshifts using BMF status, and re-fit weights against the gap outcome rather than the deficit proxies.
- ☐ Add a `fiscal_construction_null()` function so triage and dimension discovery use null eigenvalues by default.
- ☐ Naming: "degenerate" usually means a singular matrix, which only closure produces. "Construction-induced correlation" covers all five link types.

References (from memory; verify before citing).

- Pearson, K. (1897). On a form of spurious correlation which may arise when indices are used in the measurement of organs. *Proceedings of the Royal Society of London*, 60.
- Chayes, F. (1960). On correlation between variables of constant sum. *Journal of Geophysical Research*, 65.
- Chayes, F., & Kruskal, W. (1966). An approximate statistical test for correlations between proportions. *Journal of Geology*, 74.
- Horn, J. L. (1965). A rationale and test for the number of factors in factor analysis. *Psychometrika*, 30.
- Aitchison, J. (1983). Principal component analysis of compositional data. *Biometrika*, 70.
- Kronmal, R. A. (1993). Spurious correlation and the fallacy of the ratio standard revisited. *JRSS Series A*, 156.
- Higham, N. J. (2002). Computing the nearest correlation matrix. *IMA Journal of Numerical Analysis*, 22.
- Egozcue, J. J., et al. (2003). Isometric logratio transformations for compositional data analysis. *Mathematical Geology*, 35.
- Lorenzo-Seva, U., & ten Berge, J. M. F. (2006). Tucker's congruence coefficient as a meaningful index of factor similarity. *Methodology*, 2.
- Green, S. B., et al. (2012). A proposed solution to the problem with using completely random data to assess the number of factors with parallel analysis. *Educational and Psychological Measurement*, 72.
- Lovell, D., et al. (2015). Proportionality: a valid alternative to correlation for relative data. *PLoS Computational Biology*, 11.